ZENITH LEARNING / SAMPLE MATERIAL

Mathematics - Practice Paper

Original sample questions. Not an official previous-year exam paper.

1. Solve 3x + 5 = 20.

A. 3
B. 5
C. 8
D. 15

2. What is the gradient of y = 2x + 7?

A. 7
B. 2
C. 9
D. -2

3. Factorise x squared - 9.

A. (x - 3)(x + 3)
B. (x - 9)(x + 1)
C. (x - 3)(x - 3)
D. x(x - 9)

4. If f(x) = x squared + 1, find f(3).

A. 7
B. 9
C. 10
D. 12

5. A right triangle has legs 3 and 4. Find its hypotenuse.

A. 5
B. 6
C. 7
D. 8

ZENITH LEARNING / SAMPLE MATERIAL

Mathematics - Answer Guide

Original sample questions. Not an official previous-year exam paper.

1. Solve 3x + 5 = 20.

Answer: B. 5

Subtract 5 and divide by 3: x = 5.

2. What is the gradient of y = 2x + 7?

Answer: B. 2

In y = mx + c, m is the gradient.

3. Factorise x squared - 9.

Answer: A. (x - 3)(x + 3)

Use the difference of two squares.

4. If f(x) = x squared + 1, find f(3).

Answer: C. 10

Substitute 3: 9 + 1 = 10.

5. A right triangle has legs 3 and 4. Find its hypotenuse.

Answer: A. 5

Pythagoras gives the square root of 9 + 16 = 5.